

Lab sheet 13**Submission Due: 07.10.2024****Title:** Version Control with GitHub**Aims:**

- Setting up development environment
- Create repository
- Link changes to the repository

Tasks:

- Install Git on Windows
- Configure Git
- Create a GitHub Account
- Create a Local Repository
- Stage and Commit Changes
- Create a Remote Repository on GitHub
- Link Local Repository to GitHub
- Push Changes to GitHub
- Creating and Using Branches
- Pull Changes to GitHub
- Collaborating with GitHub

Activities :

1. Install Git on Windows
 - Download Git for Windows.
 - Install Git and use the default settings.
 - Verify the setup by running the following in the CMD
 - `git --version`
2. Create a GitHub Account
 - Go to GitHub and Create an account
3. Configure Git
 - Set up username and email for commits
 - `git config --global user.name "Your name"`
 - `git config --global user.email "email@example.com"`
 - Verify the configuration
 - `git config --list`
4. Create a Remote Repository on GitHub
 - Log into GitHub and create a new repository
5. Create a Local Repository
 - Open command prompt

- Navigate to your project directory
 - Initialize a new Git repository
6. Stage and Commit Changes
 - Add files to the staging area
 - `git add .` (for all files)
 - `git add file_name.txt` (for one file)
 - Commit the changes
 - `Git commit -m "Initial Commit"`
 7. Link Local Repository to GitHub
 - Copy the repository URL
 - Add the remote repository URL
 - `git remote add origin paste_url`
 8. Push Changes to GitHub
 - Push your commits to GitHub
 - `git push -u origin master`
 9. Creating and Using Branches
 - Create a new branch
 - `git checkout -b new-branch`
 - Switch to the branch
 - `git checkout new-branch`
 - Push the branch to GitHub
 10. Pull Changes from GitHub
 - To fetch and merge changes from the remote repository
 11. Collaborating with GitHub
 - Clone the repository
 - `Git clone repository_url`

Discussion:

- Basic Git Commands :
 - `git status`
 - `git log`
 - `git diff`
 - `git reset HEAD <file>`
 - `git merge <branch>`
- Conflict resolving in GitHub
- Cloning a GitHub repository